

Internship Project Proposal

1. Customer Organisation Details

Company Name:	National Dental Centre Singapore		
Contact Name:	Dinesh Ming Yang Bora	Contact No:	96638812
		E-mail:	dinesh.ming.yang.b@ndcs.com.sg

2. Project Details

*You may provide job link(s) instead.

Project Name:	AI single Tooth Orientation Prediction (AI)
Project Scope:	Please highlight which programme students you are targeting for the project. - Enterprise Business Analytics (MTECH EBAC): Data Analysis with Predictive modelling - Artificial Intelligence Systems (MTECH AIS): AI & Machine Learning & Deep Learning - Software Engineering (MTECH SE)
*Project Job Description:	The intern will build and train a machine learning model capable of classifying tooth surfaces from 3D scans or models. This involves collecting and preparing labeled dental datasets, extracting geometric and morphological features, and experimenting with suitable AI/ML techniques to achieve accurate classification. The end goal is to integrate this model into a tool that can process a given tooth geometry and label its surfaces consistently. The project will involve: Data Acquisition & Labelling – Compiling 3D tooth models and annotating surface regions with correct orientation labels. Feature Acquisition – Enhancing labelling features with geometrical features to improve AI detection rates. Model Development – Training and evaluating machine learning models Validation – Testing model accuracy on new, unseen tooth geometries. Integration & Visualization – Creating a module or UI component that highlights each orientation surface in a distinct color for easy verification.
*Tech Skills Requirements:	- Proficient in Machine learning in Python. - Proficient in Unity development
No of Intern(s)	2

Required:		
Stipend available: <i>Please provide indicative amount per month in SG\$</i>		
* Background/Aims/Objectives: <i>Provide a brief background of the project including a description of the domain and problem area. Subsequently, describe the aim and objective of the specific project for the students.</i> The objective of this project is to enhance the current dental design system by adding a web-based 2D denture design modules. The existing 3D viewer application is used to visualize jaw models, and this project will expand the system's functionality to allow users to design 2D dentures and manage the associated survey angles via an interactive user interface. The 2D designs data must be synchronized with a shared database that is also used by a windows desktop application to maintain consistency across platforms. A specific challenge will be to either integrate a DLL library to handle survey angles or develop the necessary logic themselves. This task will give the student a deep understanding of dental design software, web-based 3D/2D visualization.		
* Desired Outcome: <i>Be specific regarding the outcome of the project in terms of quantifiable and measurable deliverables unless it is a very experimental project.</i> To develop an AI-based system that automatically determines the orientation of a tooth model, identifying the lingual, buccal, mesial, and distal surfaces, which aims in downstreams workflows and improve automation/accuracy and design.		
For ISS Use Only		
Programme Name:	Project No:	Student Batch:
Accepted/Rejected/KIV:		
Students Assigned:		
Advisor Assigned:		

Note:

For MTECH Interns

Internship Period will be from mid/end March 2026 to mid/end August 2026

- **Minimum of 20 weeks (plus 1 week break in May for exam)**
- Student will be a Full-time intern, attached to sponsor companies
- Refer to Appendix A on CPF and Employment Act terms
- Refer to Appendix B on Work Pass exemption for foreign students

If you have any queries, please **contact:**

MTECH Internships: Simon Chee, simon.chee@nus.edu.sg and Pfeiffer Chung pfchung@nus.edu.sg

NUS-ISS TalentXperience

Internship Networking & Recruitment Day

Booth Requirements Form

1. Customer Organisation Details

Company Name:	National Dental Centre Singapore		
Contact Name:	Dinesh Ming Yang Bora	Contact No:	96638812
		E-mail:	dinesh.ming.yang.b@ndcs.com.sg
No of pax at booth:	2		

2. Booth Details

- Allocated booth space – 2m x 2m, inclusive of 1 x table and 2 x chairs.
- Companies will be able to supplement their space with their own collaterals and visual aids.
- Please indicate in the box below items you will be putting at your booth such as laptops, pull-up banners, TV on stand, extension wire etc.

We will be bringing:

- 1. Laptop**
- 2. Extension wire**

Do you require an electrical power point: (Please highlight)

Yes / No

Appendix A

CPF Contribution & Employment Act

CPF Contribution

According to the Central Provident Fund (CPF) policy, registered NUS students (Singaporean/PRs only) undergoing a supported internship at organisations are exempted from CPF contributions.

Source: <https://www.cpf.gov.sg/service/article/are-cpf-contributions-payable-for-interns-undergoing-an-internship-programme>

Employment Act

Source: <https://www.mom.gov.sg/faq/employment-act/as-an-intern-am-i-covered-under-the-employment-act-or-the-work-injury-compensation-act>

1. The Employment Act covers every employee who is engaged under a contract of service, including interns. This means an employer is obliged to accord statutory benefits prescribed under the Act to an intern, as long as the intern performs work and has work arrangements similar to that of a regular employee in the organisation.
2. For example, interns who enter into a contract of service should not be required to work for more than 8 hours in one day or more than 44 hours in a week. However, he may be required to work overtime if he consents to do so, subject to the limit of 12 working hours per day. Total overtime hours should also not exceed 72 hours per month. The employer must pay the intern for the extra hours of work or for working on his rest days or public holidays. The rate for overtime work should be no less than one and a half times his hourly basic rate of pay. The employer must grant the intern one rest day (which is not paid) in a week, which may be on a Sunday or any other day.
3. On the part of education institutions, they evaluate and select suitable companies for internship programmes and match students to relevant programmes. The education institutions also work with employers to brief students on the work environment, and working arrangements including whom they could approach to provide feedback or seek guidance. Student interns should be clear about their working arrangements before accepting and

starting the internship. Education institutions should make a final assessment of the student's internship performance, taking into account the feedback from both the student and the employer. Both the education institution and employer are expected to provide regular feedback to students throughout the internship. Students can also provide their feedback about the internship to their education institutions, so that the institutions and employers can continually think of ways to improve the internship experience.

Appendix B

Work pass exemption for foreign students

The NUS-ISS Master of Technology student internship is an integral part of the degree coursework. The internship itself constitutes a graded module (ISY5007, EBA5007 and SWE5007) and contributes toward the student's graduation requirement. Therefore, foreign students enrolled in the full-time programme are exempted from work pass requirements only during their internship from 01 March 2026 to 30 Sep 2026.2025

Source: <http://www.mom.gov.sg/passes-and-permits/work-pass-exemption-for-foreign-students>